When Does Density Beat Centroids? A Comparative Study of Clustering Methods Across Data Geometries

[**DECIDE**: final title. Proposed: "When Clustering Fails: Which Data Break Which Methods, and Can You Tell Without Labels?", to match the research question]

SC4020 Project 1 — Group XX

[**All**: full names exactly as on student cards; replace XX with group ID]

Abstract

Clustering methods differ in what they take a cluster to be, and real data break those ideas in different ways. We compare K-means, K-means++, DBSCAN, HDBSCAN, Gaussian mixtures and spectral clustering on seven datasets, each chosen to test one data property: curved shapes, long belts of uneven density, a dense city with no gaps, rare classes, high dimensions, and data with no gaps between groups. Every method is tuned by one label-free rule. We ask where each method fails, whether the scores available without labels reveal those failures, and whether a method built for the property repairs them. Each method failed where the data broke its idea of a cluster. K-means cut the curved classes of make_moons (ARI 0.48 against 1.00 for spectral clustering); one DBSCAN radius chained the dense western Pacific into one cluster of 36% of earthquakes while HDBSCAN followed the plate belts; and in 50 dimensions the density methods fell to ARI 0.03 or below. The silhouette did not reveal these failures: on all four labelled datasets the method with the top silhouette was not the one that matched the labels best, and on make_moons it chose a DBSCAN setting that breaks the moons into 72 fragments (ARI 0.01). Rerunning each result on shuffled data caught methods that found nothing real, but not one dominant cluster or the wrong structure. No method produced meaningful clusters in data with no gaps. The choice of method should follow from the structure of the data, and every score needs a check against fake data and against the share of the largest cluster.

[**DECIDE**: Thesis, proposed 7 Oct (Wei Yew), to match the research question in Section 1: each method fails where the data break its idea of a cluster (RQ1); the silhouette does not reveal these failures, and a fake-data check reveals only some (RQ2); a method built for the property fixes some failures, and none fixes data with no gaps (RQ3).]

1 Introduction

1.1 Motivation

Clustering is used where no answer key exists. A seismologist grouping earthquake epicentres wants active fault systems; a housing analyst grouping resale transactions wants market segments. Neither has labels to check a result against.

Each clustering method carries its own idea of what a cluster is. K-means [12] looks for compact groups around a centre and assigns every point to one of them. DBSCAN [8] and HDBSCAN [5] look for dense regions separated by sparse space, and leave points in the sparse space as noise. Whether or not that idea matches the data, the method returns a partition. On one year of global earthquakes, K-means returns six tidy regions, each joining several plate boundaries that have nothing to do with each other (Section 3.4.2). Nothing in the output says that this has happened. An analyst without labels usually judges the result by an internal score such as the silhouette [16], which rates how compact and well separated the clusters are, and that score is itself built on the K-means idea of a cluster.

1.2 Research question

This report asks one question: *when a clustering method's idea of a cluster does not match the data, does it fail, and can the failure be detected without labels?* We split it into three parts.

RQ1, cause. Which property of the data breaks which method? We consider cluster shape, density that varies between clusters, the number of dimensions, the absence of gaps between groups, and rare classes.

RQ2, detection. Do checks that need no labels reveal those failures? We test the silhouette, the Davies–Bouldin index, the share of points labelled noise, and a fake-data check: the same method run on data whose features have been shuffled independently, which keeps each feature's values but removes any joint structure.

RQ3, remedy. For each failure, does a method built for that property help, and what does it cost?

We compare six methods, K-means with random starts, K-means++ [2], DBSCAN, HDBSCAN, Gaussian mixtures (GMM) and spectral clustering, on seven datasets (Table 2), all run the same way. Each dataset was chosen to test one data property, most of them a property that breaks at least one method's assumption. Four have class labels, so a method's output can be scored against them with the Adjusted Rand Index (ARI) [11], and the internal scores can be checked against that answer. Three have no labels (earthquakes, Uber pickups, HDB resale flats); there we judge failures on maps and plots, against known geography.

1.3 Findings in brief

RQ1. Each method failed where the data broke its assumption. K-means and GMM cut the two interleaved half-moons of D1 wrongly (ARI 0.48 and 0.50, against 1.00 for spectral clustering) and drew regions that join unrelated plate boundaries on D2. With one radius, DBSCAN chained the dense western Pacific into one cluster holding 36% of all earthquakes, while HDBSCAN followed the belts. In 50 dimensions (D5) both density methods barely matched the labels (ARI 0.03 or below). On D3 and D8, where there are no gaps between groups, both density methods returned one dominant cluster (92% to 99.7%).

RQ2. The silhouette did not reveal these failures. On all four labelled datasets, the method with the highest silhouette was not the one that matched the labels best. On D1 it even chose the failure: the rule's DBSCAN setting breaks the moons into 72 fragments (ARI 0.009) because they score a higher silhouette than the setting that recovers the moons exactly. On Ecoli (D4) HDBSCAN puts 97% of proteins in one cluster and has the highest silhouette. The fake-data check does better, but only partly. It flags results with no real structure: D1 DBSCAN scores 0.523 against 0.822 on shuffled data, and D4 HDBSCAN no better than shuffled data (0.559 against 0.553). It does not flag one dominant cluster (D8, where both density methods beat shuffled data) or the wrong structure (D2, where K-means beats shuffled data while joining unrelated belts).

RQ3. Methods built for the property that broke another method repaired some failures and not others. Spectral clustering recovered the curved moons that K-means cut; HDBSCAN followed belts of uneven density that one DBSCAN radius could not; GMM, with a covariance per component, matched compact classes better than K-means (ARI 0.618 against 0.508 on D4, 0.915 against 0.897 on D6). Nothing produced meaningful density clusters in data with no gaps, or in 50 dimensions.

Section 2 describes the methods and Section 3 the datasets, protocol and results, grouped by data property. Section 4 turns them into guidance.

1.4 Related work

Centroid and mixture models. Lloyd's algorithm [12] minimises the within-cluster sum of squared distances by alternating assignment and centroid updates. It converges to a local optimum

that depends on the starting centres. K-means++ [2] picks each starting centre with probability proportional to its squared distance from the centres already chosen, which gives an $O(\log k)$ approximation to the optimum in expectation. A Gaussian mixture fitted by EM [7] generalises K-means to ellipsoidal clusters with soft assignments.

Density-based methods. DBSCAN [8] defines a cluster as a maximal set of density-connected points. A point is a core point if at least `minPts` points lie within radius ε. Points reachable from no core point are noise. The method needs no k and imposes no shape, but its single global ε assumes all clusters have similar density. OPTICS [1] orders points by reachability so that clusters at several densities can be read from one plot. HDBSCAN [5] builds a hierarchy over mutual reachability distances and keeps the clusters that persist longest. This removes ε and leaves a minimum cluster size as the main parameter.

Graph-based methods. Spectral clustering [14] embeds the points using the leading eigenvectors of a normalised affinity matrix, then runs K-means in that embedding. Groups connected through the affinity graph become compact in the embedding, so non-convex clusters can be separated. A dense affinity matrix needs $O(n^2)$ memory, which limits the method to modest n.

Validation and cluster tendency. Internal indices, such as the silhouette [16], Davies–Bouldin [6] and Calinski–Harabasz [4] indices, score a partition from the data alone. External indices such as the Adjusted Rand Index [11] compare it with known labels. Von Luxburg et al. [13] argue that no criterion can judge a clustering apart from its intended use. A separate line of work tests whether data contain clusters at all. The Hopkins statistic [9] compares nearest-neighbour distances in the data with those in uniformly random points. The gap statistic [17] compares within-cluster dispersion with its value on reference data drawn from a null distribution. Our fake-data check applies the same idea to every method we compare. In high dimensions the ratio of nearest to farthest neighbour distance tends to one [3], which weakens any method built on a density threshold.

2 Methods

2.1 K-Means and K-Means++

How it works. K-means [12] finds k centres that minimise the within-cluster sum of squares $J = \sum_i \min_j \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$: assign each point to its nearest centre, move each centre to the mean of its points, repeat. J never rises, so it stops at a local optimum that depends on the start. Random starts pick k points uniformly; K-means++ [2] picks each next centre with probability proportional to its squared distance from the nearest chosen one. One iteration costs $O(nkd)$.

Pros. Fast and simple; scales to large n; K-means++ makes a good optimum more likely.

Cons. Assumes compact, round clusters of similar spread: cuts curved or elongated ones. Needs k, assigns every point (no noise), and depends on feature scale.

Our implementation. `kmeans` (K-means++) and `kmeans_random_init` wrap scikit-learn's `KMeans` with `n_init`=10 and seed 42. Parameter assessed: $k \in \{2, \ldots, 10\}$ (a wider grid up to 50 on D2), chosen by `select_kmeans_k_by_null_gap`: among k stable across 10 seeds (pairwise ARI ≥ 0.95), the largest silhouette gap over column-shuffled data. On labelled data k is the number of classes.

2.2 DBSCAN

How it works. DBSCAN [8] takes a radius ε and a count `minPts`. A point with at least `minPts` neighbours within ε is a core point; core points within ε of each other form a cluster, points near a core point join it as border points, and the rest are noise. With a spatial index a fit costs about $O(n \log n)$ in few dimensions, up to $O(n^2)$.

Pros. No k; any cluster shape; labels noise; on coordinates ε is a distance in km.

Cons. One radius for all clusters: sparse clusters become noise or dense ones merge. Needs gaps between clusters, and weakens in many dimensions, where distances become alike [3].

Our implementation. `dbscan` wraps scikit-learn's `DBSCAN` with Euclidean distance (on D2 between unit vectors, which follows the sphere). Parameters assessed: `min_samples` around $\text{clip}(2d, 4, 20)$ (half, once, twice and four times it) and ε at 0.5 to 4 times the knee of the k-distance curve (`k_distance_curve`, `find_knee`), chosen by `select_dbscan_by_shared_rule`: among settings with at least 2 clusters and at most 50% noise, the highest silhouette.

2.3 HDBSCAN

How it works. HDBSCAN [5] removes ε. With the core distance of a point (to its m-th neighbour), the mutual reachability distance $d_{\text{mr}}(a, b) = \max\{\text{core}(a), \text{core}(b), d(a, b)\}$ pushes sparse points apart. Its minimum spanning tree holds DBSCAN's clusterings at every density; splits smaller than `min_cluster_size` are dropped and the clusters that persist longest are kept, the rest is noise. A fit costs between $O(n \log n)$ and $O(n^2)$.

Pros. No k or ε; clusters of different density in one run; labels noise.

Cons. `min_cluster_size` is a count, so results change with sample size; can call large parts of the data noise; finds nothing where there are no density gaps.

Our implementation. `hdbscan` wraps scikit-learn's `HDBSCAN`. Parameters assessed: `min_cluster_size` $\in \{5, 10, 15, 20, 30, 50, 100, 200, 400\}$ up to $n/10$, and `min_samples` $\in$ {default, 5, 10}, chosen by `select_hdbscan_by_shared_rule`.

2.4 Gaussian Mixture Model

How it works. A Gaussian mixture model (GMM) models the data as $p(\mathbf{x}) = \sum_{j=1}^{k} \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$, fitted by expectation–maximisation [7]: the E-step computes each point's probability under every component, the M-step re-estimates weights, means and covariances. Each point goes to its most probable component. One iteration with full covariances costs $O(nkd^2 + kd^3)$.

Pros. Ellipsoidal clusters of different size, orientation and weight; soft assignments; BIC gives a principled way to choose k.

Cons. Each cluster must be one convex blob: fails on curved or nested shapes. Needs k, has no noise label, and full covariances need many points per component when d is large.

Our implementation. `gmm` wraps scikit-learn's `GaussianMixture` [15] with `n_init`=3 and seed 42. Parameters assessed: k over the same grid as K-means and `covariance_type` $\in$ {full, tied, diag, spherical}, chosen by lowest BIC (`select_gmm_by_bic`); on labelled data k is the number of classes.

2.5 Spectral clustering

How it works. Spectral clustering [14] builds an affinity matrix between points, takes the k leading eigenvectors of its normalised graph Laplacian as a new embedding, and runs K-means there. Points linked by chains of close neighbours land together, even if far apart in the input. A dense affinity costs $O(n^2)$ memory and up to $O(n^3)$ time; a sparse neighbour graph is cheaper.

Pros. Recovers curved and non-convex clusters; a neighbour graph adapts to local scale.

Cons. Needs k; no noise label; sensitive to the affinity; quadratic cost limits n.

Our implementation. `spectral` wraps scikit-learn's `SpectralClustering` (seed 42). Affinity fixed per dataset: RBF with $\gamma = 1$ on D6, a 10-nearest-neighbour graph elsewhere. k is taken from the K-means selection, not tuned. The dense RBF affinity is skipped above 10,000 points; the sparse neighbour graph runs at any size.

Table 1: Structural assumptions of the methods compared.

	K-Means	K-Means++	DBSCAN	HDBSCAN	GMM	Spectral
Cluster shape assumed	isotropic	isotropic	arbitrary	arbitrary	ellipsoidal	connected (graph)
Requires k	yes	yes	no	no	yes	yes
Explicit noise label	no	no	yes	yes	no	no
Handles varying density	no	no	no	yes	partly[b]	partly[c]
Parameter is interpretable	no	no	yes (km)	partly	no	no
Time complexity	$O(nkdi)$	$O(nkdi)$	$O(n \log n)$[a]	$O(n \log n)$[a]	$O(nkd^2i)$	$O(n^2d + n^3)$[d]

[a] With an effective spatial index; $O(n^2)$ worst case. [b] Per-component covariance and weight. [c] With a k-NN affinity, which adapts to local scale. [d] Dense RBF affinity and full eigendecomposition; lower with a sparse k-NN graph.

3 Experiments

3.1 Datasets

3.2 Preprocessing

- **Latitude, longitude (D2)** → 3D points on the unit sphere, so straight-line distance follows the Earth's surface and the 180° meridian is no border.
- **Latitude, longitude (D3)** → kilometres, projected about the mean latitude.
- **Features in raw units (D1, D4, D6, D8)** → mean 0, standard deviation 1.
- **Images (D5)** → pixels scaled to $[0, 1]$, then 50 principal components.
- **Stepped values (D8)** → uniform jitter of up to half a step, so tied values do not form artificial dense spots: ± 1.5 storeys, $\pm 0.5\,\mathrm{m}^2$ floor area, $\pm \frac{1}{24}$ year of lease.

3.3 Evaluation metrics and parameter selection

Metrics. Every result table reports the same columns. *Internal* scores need no labels. The silhouette coefficient [16] of a point is $(b - a)/\max(a, b)$, where a is its mean distance to its own cluster and b its mean distance to the nearest other cluster; we report the mean over points, from -1 to 1, higher is better. The Davies–Bouldin index [6] averages, over clusters, the worst ratio of within-cluster spread to between-centre distance; lower is better. Both are computed on clustered points only, so they say nothing about the points a density method calls noise. Two *coverage* figures therefore stand next to them: the noise fraction and the share of points in the largest cluster. *External* scores need labels. The Adjusted Rand Index (ARI) [11] counts pairs of points on which a

Table 2: Datasets and why each was chosen.

ID	Dataset	n	d	Labels	Preparation	Why chosen
D1	make_moons [15]	1,000	2	2 classes	noise 0.07; standardised	Non-convex moons: shows where density and spectral methods beat K-means.
D2	Earthquakes 2023, M≥4.5 [19]	7,643	3	none	8 eruptions dropped; epicentre as unit vector	Long belts of uneven density: exposes K-means (compact clusters) and DBSCAN (one radius).
D3	Uber NYC, Apr 2014 [23]	15,000	2	none	sampled from 558,019 in NYC box; km	Dense city with no gaps: shows density methods collapse into one cluster.
D4	Ecoli [22]	336	7	8 sites	standardised	Seven dimensions, tiny classes: favours K-means; density methods lose rare classes.
D5	Fashion-MNIST [18]	3,000	50	10 classes	stratified; PCA 784→50 (86.5% var.)	50 dimensions, overlapping classes: exposes density methods in high dimensions.
D6	Wine [21]	178	13	3 cultivars	standardised	Compact classes, no gaps: favours K-means; shows the silhouette misleads.
D8	HDB resale 2017–2026 [20]	20,000[a]	4	none	price index-adjusted; ties jittered; standardised	No gaps between groups: exposes density methods, which need sparse space to cut along.

[a] Random sample (seed 0) of the 242,634 sales from January 2017 in the download of 13 Oct 2026.

clustering and the labels agree, corrected so that a random clustering scores about 0 and a perfect one 1. Normalised mutual information (NMI) measures the same agreement in terms of shared information. For both, noise counts as one more cluster, so a method cannot raise its score by discarding hard points. We also report the fit time.

Which metric judges which dataset. No single score suits every dataset, so before clustering we measured each dataset's shape (Table 3). Density spread is the 90th over the 10th percentile of each point's distance to its 10th neighbour (1 means even density). Far outliers are points whose distance lies beyond $Q_3 + 3\,\mathrm{IQR}$; this share is the noise a dataset can be expected to have, so a noise fraction well above it means a method discarded real points rather than removed outliers. With labels we also score the true classes as if they were a clustering, and compare two classifiers: 1-nearest-neighbour follows any shape, nearest-centroid only convex, roughly spherical classes, so a 1-NN accuracy well above the nearest-centroid accuracy means the classes are not convex. Three rules follow. (i) On labelled data, ARI ranks the methods and NMI confirms it. (ii) The silhouette and Davies–Bouldin assume compact, convex clusters. They are trusted only when the true classes have that shape (D6), and are never trusted when the true classes themselves score worse than a wrong clustering (D1: 0.387 for the moons against 0.495 for K-means). (iii) On unlabelled data they are the only scores we have, so they are read only together with the fake-data silhouette, the noise fraction against the far-outlier share, and the largest cluster.

Fake-data check. An internal score has no fixed scale: a silhouette of 0.3 can be strong on one dataset and meaningless on another. Each chosen setting is therefore run again on fake data that keeps each feature's distribution but has no joint structure: every column shuffled independently (seed 0), on every dataset. A result counts as structure only if the real data score clearly above the

Table 3: Shape of each dataset and the metrics used to judge methods on it. Spread: 10-NN distance, 90th over 10th percentile. Outliers: far-outlier share. S_{true}, DB_{true}: silhouette and Davies–Bouldin of the true classes. 1-NN / NC: cross-validated accuracy of 1-nearest-neighbour and nearest-centroid.

ID	Measured shape	Reading	Ranked by	Read alongside
D1	2 classes, 50%/50%; S_{true} 0.387, DB_{true} 1.02; 1-NN 1.00 / NC 0.86; spread 2.1; outliers 1.4%	Separated but not convex: S and DB score the true moons below K-means.	ARI	NMI; noise vs. 1.4%
D2	no labels; spread 34.6; outliers 3.8%	Long belts of very uneven density: S and DB favour compact blobs, not belts.	S, DB	null S; largest cluster; noise vs. 3.8%
D3	no labels; spread 8.5; outliers 9.2%	One dense city with no gaps and a sparse fringe: one giant cluster scores a high S.	S, DB	null S; largest cluster; noise vs. 9.2%
D4	8 classes, 0.6% to 42.6%; S_{true} 0.226, DB_{true} 1.59; 1-NN 0.76 / NC 0.82; spread 2.3; outliers 4.2%	Roughly convex but overlapping, with tiny classes: ARI is driven by the large classes.	ARI	NMI; noise vs. 4.2%; largest cluster
D5	10 classes, 10% each; S_{true} 0.064, DB_{true} 3.01; 1-NN 0.78 / NC 0.68; spread 2.0; outliers 0.0%	Heavily overlapping, even density: true classes score S near 0, so S cannot rank.	ARI	NMI; any noise is lost data
D6	3 classes, 27% to 40%; S_{true} 0.280, DB_{true} 1.41; 1-NN 0.95 / NC 0.97; spread 1.7; outliers 2.2%	Compact, convex, even density: the one case where S and DB match the classes.	ARI	S and DB at equal noise; noise vs. 2.2%
D8	no labels; spread 3.4; outliers 1.2%	No gaps and fairly even density: little to remove as noise.	S, DB	null S; largest cluster; noise vs. 1.2%

fake data.

Selection rule. One rule picks the setting of DBSCAN and HDBSCAN on every dataset: among the settings that give at least 2 clusters and label at most 50% of points as noise, take the one with the highest silhouette. The noise cap is needed because the silhouette ignores noise. Without it, the silhouette prefers settings that throw most of the data away. On D1 it would pick DBSCAN with $\varepsilon = 0.089$, which scores 0.948 by calling 96.5% of points noise; on D5, DBSCAN with $\varepsilon = 2.94$, which scores 0.525 with 90.4% noise. For K-means, K is the number of classes on labelled data; otherwise, among the K whose 10 seeds agree (pairwise ARI ≥ 0.95), the one with the largest gap between the real silhouette and the mean silhouette on 3 fake copies. GMM takes the components and covariance type with the lowest BIC, with the number of components fixed to the number of classes on labelled data. Spectral takes the K of K-means. On labelled data we also record the setting with the best ARI, so the cost of choosing without labels is visible.

Parameter grids. The same grids are used on every dataset. K-means and K-means++: $K = 2$ to 10 with 10 starts (on D2 a coarse grid from 2 to 50, since a global catalogue has dozens of belts). GMM: the same K grid and four covariance types (full, tied, diagonal, spherical). DBSCAN: `minPts` at half, once, twice and four times $m = \text{clip}(2d, 4, 20)$, and ε at 0.5 to 4 times the knee of the k-distance curve. HDBSCAN: `min_cluster_size` in $\{5, 10, 15, 20, 30, 50, 100, 200, 400\}$ up to $n/10$, and `min_samples` in {default, 5, 10}. Spectral: an RBF affinity on D6 and a 10-nearest-neighbour graph elsewhere. Silhouettes are estimated on a sample of 5,000 points when n is larger.

Reproducibility. Every method uses seed 42 and the fake data seed 0. All seven notebooks ran on Python 3.13.15, scikit-learn [15] 1.9.1, NumPy 2.5.3 and pandas 3.0.5, on an AMD Ryzen 7 5800H with 16 GB of memory.

3.4 Comparison by data property

Table 4: Labelled datasets at the settings chosen without labels (Section 3.3). S: silhouette; DB: Davies–Bouldin, both on clustered points only; null S: silhouette of the same setting on fake data (every column shuffled; "none": no clusters formed). Bold: best S and best ARI per dataset.

Data	Method (setting)	Clusters	Noise	Largest	S	DB	ARI	NMI	Time (s)	Null S
D1	K-Means ($K = 2$)	2	0.0%	50.5%	0.495	0.812	0.481	0.383	0.006	0.353
D1	K-Means++ ($K = 2$)	2	0.0%	50.5%	0.495	0.812	0.481	0.383	0.007	0.353
D1	DBSCAN (0.0486, 4)	72	47.1%	1.8%	0.523	0.565	0.009	0.205	0.003	0.822
D1	HDBSCAN (100)	2	15.6%	42.4%	**0.529**	0.735	0.709	0.676	0.018	none
D1	GMM (2, tied)	2	0.0%	50.0%	0.495	0.812	0.501	0.400	0.008	0.353
D1	Spectral ($K = 2$, 10-NN)	2	0.0%	50.0%	0.387	1.023	**1.000**	1.000	0.014	0.319
D4	K-Means ($K = 8$)	8	0.0%	20.8%	0.236	1.135	0.379	0.585	0.009	0.192
D4	K-Means++ ($K = 8$)	8	0.0%	23.8%	0.260	1.091	0.508	0.641	0.012	0.187
D4	DBSCAN (0.809, 7)	2	31.5%	46.1%	0.471	0.833	0.395	0.415	0.002	0.365
D4	HDBSCAN (5)	2	0.3%	97.0%	**0.559**	0.637	0.038	0.118	0.003	0.553
D4	GMM (8, full)	8	0.0%	38.4%	0.157	2.024	**0.618**	0.597	0.039	0.149
D4	Spectral ($K = 8$, 10-NN)	8	0.0%	18.5%	0.231	1.305	0.392	0.591	0.019	0.156
D5	K-Means ($K = 10$)	10	0.0%	16.0%	0.202	1.470	0.344	0.512	0.050	0.081
D5	K-Means++ ($K = 10$)	10	0.0%	16.0%	**0.202**	1.470	0.343	0.512	0.066	0.083
D5	DBSCAN (4.42, 40)	2	22.7%	75.7%	0.138	1.211	0.028	0.103	0.011	none
D5	HDBSCAN (15, 5)	2	32.9%	66.3%	0.039	1.754	0.023	0.064	0.330	0.229
D5	GMM (10, full)	10	0.0%	18.3%	0.129	2.497	0.389	0.572	14.715	0.049
D5	Spectral ($K = 10$, 10-NN)	10	0.0%	20.0%	0.194	1.568	**0.398**	0.586	0.136	0.074
D6	K-Means ($K = 3$)	3	0.0%	36.5%	0.285	1.389	0.897	0.876	0.005	0.066
D6	K-Means++ ($K = 3$)	3	0.0%	36.5%	0.285	1.389	0.897	0.876	0.006	0.066
D6	DBSCAN (3.09, 40)	2	23.6%	53.4%	**0.365**	1.080	0.431	0.536	0.001	none
D6	HDBSCAN (5, 10)	2	42.1%	42.1%	0.363	1.046	0.260	0.354	0.002	none
D6	GMM (3, diagonal)	3	0.0%	39.3%	0.280	1.412	**0.915**	0.892	0.007	0.054
D6	Spectral ($K = 3$, RBF)	3	0.0%	68.0%	0.249	1.271	0.445	0.572	0.009	0.011

3.4.1 Curved clusters

On D1 the methods that assume convex clusters fail and the graph method succeeds (Figure 1). K-Means++ (ARI 0.481) and GMM (0.501) draw a nearly straight boundary that cuts each moon in two. Spectral clustering on a 10-nearest-neighbour graph recovers both moons exactly (ARI 1.000), because each moon is a chain of close neighbours with a gap wider than any link between the moons. HDBSCAN finds both moons too but calls 15.6% of points noise, mostly the sparse moon tips, so its ARI is 0.709. DBSCAN fails, and not because of its assumptions: its grid contains a setting that recovers the moons exactly ($\varepsilon = 0.229$, `minPts` 3, ARI 1.000), but the selection rule prefers $\varepsilon = 0.049$, which breaks the moons into 72 fragments with 47.1% noise (ARI 0.009), because the fragments score a higher silhouette (0.523 against 0.387).

3.4.2 Long belts of uneven density

On D2 clusters should follow plate boundaries: long, narrow belts whose density varies along and between belts. HDBSCAN, which sets a density level per cluster, returns 28 clusters that follow the belts (largest 7.5% of earthquakes) and calls 30.7% noise; its silhouette is the highest on D2 (0.756, against 0.317 on fake data). DBSCAN with one radius (≈250 km) must trade the sparse belts against the dense ones: it keeps more earthquakes (11.3% noise) but chains the dense western Pacific into one cluster holding 35.7% of all earthquakes. K-Means++ and Spectral ($K = 6$) cover the globe with six regions, each joining several unrelated belts (Figure 2); their silhouettes (0.516 and 0.334) still beat fake data (0.319 and 0.295), so the score confirms that the data are not random, not that the clusters are meaningful. GMM, chosen by BIC, uses all 50 components the grid allows: BIC keeps rewarding smaller ellipses along the belts, so it gives no natural number of clusters.

Table 5: Unlabelled datasets, same columns as Table 4 without ARI and NMI. D2 DBSCAN radius converted from unit-vector distance to km on the Earth's surface.

Data	Method (setting)	Clusters	Noise	Largest	S	DB	Time (s)	Null S
D2	K-Means ($K = 6$)	6	0.0%	31.3%	0.512	0.728	0.023	0.320
D2	K-Means++ ($K = 6$)	6	0.0%	31.0%	0.516	0.740	0.026	0.319
D2	DBSCAN (250 km, 12)	42	11.3%	35.7%	0.420	0.374	0.034	0.599
D2	HDBSCAN (50)	28	30.7%	7.5%	**0.756**	0.329	0.232	0.317
D2	GMM (50, full)	50	0.0%	7.3%	0.554	0.631	1.934	-0.135
D2	Spectral ($K = 6$, 10-NN)	6	0.0%	31.6%	0.334	0.914	0.118	0.295
D3	K-Means ($K = 5$)	5	0.0%	44.5%	0.481	0.699	0.025	0.396
D3	K-Means++ ($K = 5$)	5	0.0%	44.5%	0.481	0.699	0.893	0.338
D3	DBSCAN (2.81 km, 16)	2	0.2%	97.7%	**0.757**	0.186	0.903	none
D3	HDBSCAN (200)	3	4.6%	91.5%	0.638	0.267	0.975	0.660
D3	GMM (9, full)	9	0.0%	25.6%	0.359	1.624	0.804	0.331
D3	Spectral ($K = 5$, 10-NN)	5	0.0%	35.3%	0.157	1.389	1.070	0.308
D8	K-Means ($K = 4$)	4	0.0%	35.4%	0.317	1.125	0.050	0.216
D8	K-Means++ ($K = 4$)	4	0.0%	35.6%	0.318	1.125	0.898	0.216
D8	DBSCAN (0.614, 4)	3	0.2%	99.7%	0.438	0.436	0.567	0.189
D8	HDBSCAN (5)	2	0.3%	99.7%	**0.496**	0.406	1.596	0.136
D8	GMM (10, full)	10	0.0%	15.8%	0.176	1.537	1.456	0.088
D8	Spectral ($K = 4$, 10-NN)	4	0.0%	67.1%	0.205	1.227	2.413	0.167

3.4.3 A dense city with no gaps

D3 has density contrast but no gaps (Figure 3): Manhattan, Brooklyn and Queens meet without empty space, and only the airports stand apart. Both density methods therefore keep the city together. DBSCAN returns the city (97.7% of pickups) and JFK; HDBSCAN returns the city (91.5%), JFK and LaGuardia with 4.6% noise. DBSCAN has the top silhouette on D3 (0.757) because two well-separated groups score well whatever their sizes. HDBSCAN scores below its setting on fake data (0.638 against 0.660), and so does Spectral (0.157 against 0.308). The k-based methods divide the core: K-Means++ into five regions (silhouette 0.481 against 0.338 on fake data) and GMM into nine. Their borders follow their objectives, not gaps in the data.

3.4.4 Rare classes in seven dimensions

D4 has eight localisation sites, three of them with 2 to 5 proteins, in seven dimensions with no density gaps between classes. The methods that model compact groups do best: GMM reaches ARI 0.618, K-Means++ 0.508 and Spectral 0.392 at $K = 8$. DBSCAN reaches 0.395 only by calling 31.5% of proteins noise. HDBSCAN puts 97.0% of proteins in one cluster (ARI 0.038), yet has the highest silhouette on D4 (0.559), no better than the same setting on fake data (0.553). With the labels the gap narrows only for the k-based methods: GMM reaches 0.741 (7 spherical components) and K-Means++ 0.731 ($K = 7$), against 0.398 for HDBSCAN (29.8% noise) and 0.395 for DBSCAN.

3.4.5 High dimensions

On D5 ($d = 50$) the k-based methods do best: Spectral (ARI 0.398), GMM (0.389) and K-Means++ (0.343), all well short of the labels because shirts, T-shirts, pullovers and coats overlap. Both density methods collapse to two clusters: DBSCAN puts 75.7% of images in one and 22.7% in noise (ARI 0.028), HDBSCAN 66.3% in one and 32.9% in noise (ARI 0.023). This is not a bad choice of setting: the best density settings in the grid reach only 0.034 and 0.066. In 50 dimensions the distances to the k-th neighbour are similar for every point, so no single density threshold separates the overlapping classes. HDBSCAN's silhouette (0.039) is below its setting on fake data (0.229).

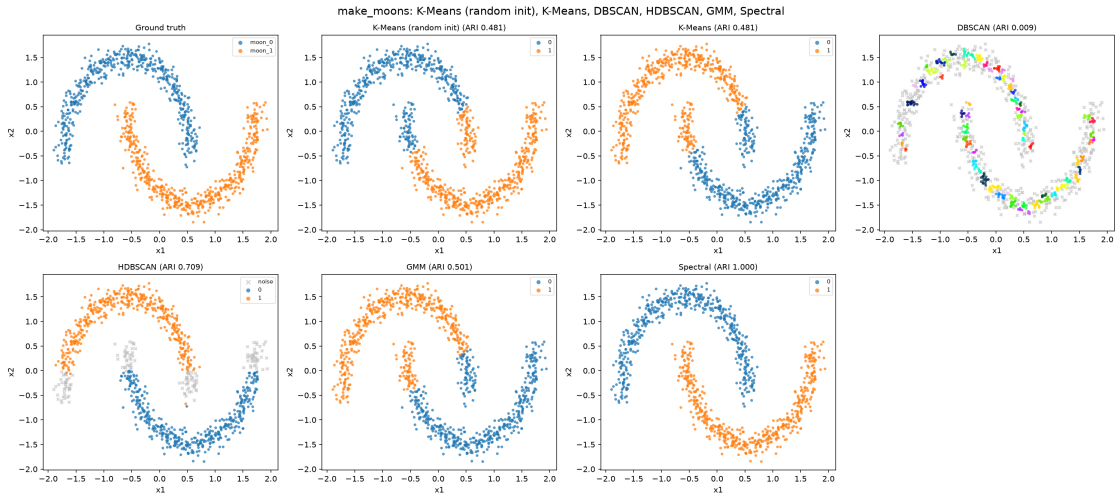


Figure 1: D1 make_moons at the chosen settings. K-means and GMM cut both moons with a straight boundary; Spectral follows the curves (ARI 1.000); HDBSCAN loses the tips to noise; DBSCAN at the radius the silhouette prefers breaks the moons into 72 fragments. Grey crosses: noise.

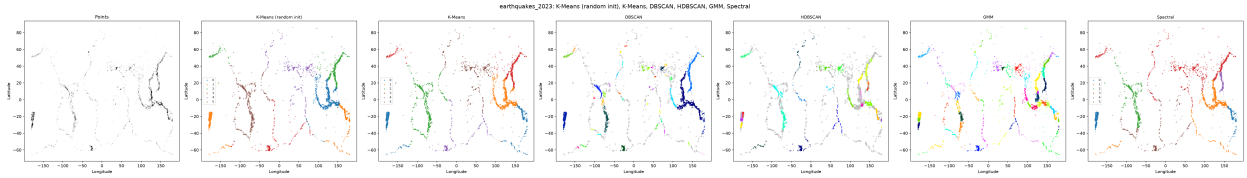


Figure 2: D2 at the chosen settings (grey: noise). HDBSCAN follows the plate belts; DBSCAN joins the dense western Pacific into one cluster; K-means and Spectral draw six regions that each span several belts; GMM splits the belts into 50 pieces.

3.4.6 No gaps between groups

D8 is one continuous mass: flats come in every combination of size, lease, storey and price. Table 5 shows what each method does with it (Figure 4). The density methods have no sparse region to cut along: DBSCAN and HDBSCAN put 99.7% of sales in one cluster and leave fewer than 100 outlying sales outside it. Their silhouettes (0.438 and 0.496) are the highest on D8 and well above fake data (0.189 and 0.136), because a giant cluster plus a tiny distant one scores well. K-means finds four segments with a silhouette of 0.318 against 0.216 on fake data, a weak but real partition; Spectral leaves 67.1% of sales in one of its four clusters, and GMM spreads ten components over the mass. On data with no gaps a density method has nothing to find, and neither the silhouette nor the fake-data check says so; only the share of the largest cluster does.

3.4.7 Silhouette versus ground truth

On every labelled dataset the method with the highest silhouette is not the one that matches the labels best (Table 4): on D1 HDBSCAN (0.529, ARI 0.709) against Spectral (ARI 1.000); on D4 HDBSCAN (0.559, ARI 0.038) against GMM (0.618); on D5 K-Means++ (0.202, ARI 0.343) against Spectral (0.398); on D6 DBSCAN (0.365, ARI 0.431) against GMM (0.915). Over the five methods the rank correlation between silhouette and ARI is -0.30 on D1, -0.70 on D4, -0.60 on D6 and 0.50 on D5, the only dataset with no density method in the lead. Two mechanisms recur. The silhouette ignores noise, so a density method can raise it by discarding hard points (D6 DBSCAN, 23.6% noise). It also rewards one giant cluster plus a small distant one (D4 HDBSCAN, 97.0% in one cluster).

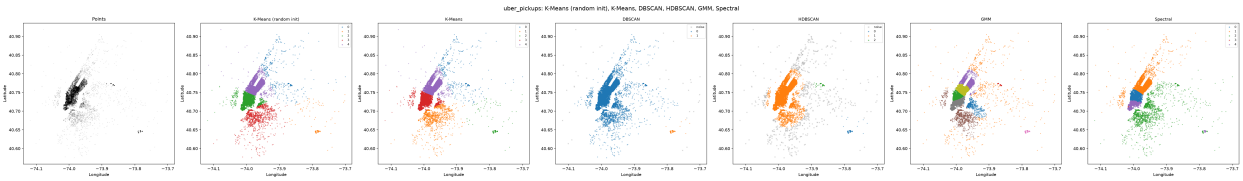


Figure 3: D3 at the chosen settings (grey: noise). DBSCAN and HDBSCAN keep the city as one cluster and separate only the airports; K-means, GMM and Spectral divide the core along borders of their own making. Cause: no gaps inside the city.

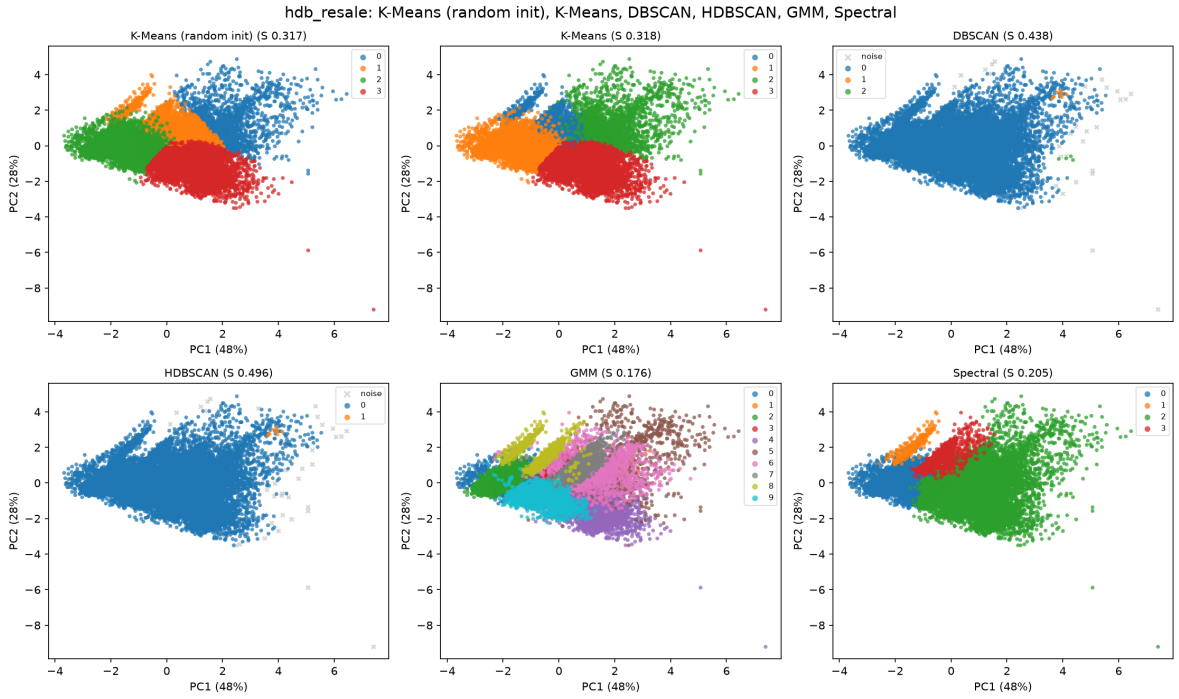


Figure 4: D8 at the chosen settings, drawn on the first two principal components (grey: noise). K-means cuts the mass into four segments; DBSCAN and HDBSCAN return one cluster holding 99.7% of sales. Cause: no gaps between groups.

D6 Wine and the fake-data check. **[HoangAnh]** On D6, DBSCAN and HDBSCAN pass the noise cap with 23.6% and 42.1% of the wines as noise, take the two highest silhouettes (0.365 and 0.363) and the two lowest ARI (0.431 and 0.260), while GMM and K-means reach 0.915 and 0.897 with silhouettes near 0.28. The fake-data check does not separate good from bad here either. Every method that assigns all points scores far above its shuffled-column silhouette (0.285 against 0.066 for K-means), and the density methods form no cluster at all on shuffled data. So each result passes the check, including the density results that match the labels worst: what they find, a dense core of one or two cultivars, is real structure, but it is not the class structure. The check shows that a result is not an artefact of the marginals. It does not show which of several real structures a method has found.

What the fake-data check catches. Across the seven datasets a setting scores no better than on fake data in six cases: DBSCAN on D1 (0.523 against 0.822) and D2 (0.420 against 0.599), HDBSCAN on D3, D4 and D5, and Spectral on D3. The three on labelled data are the clear failures by ARI (D1 DBSCAN, D4 and D5 HDBSCAN), so the check catches results with no real structure. The check misses the D8 density results, which beat fake data while holding 99.7% of sales in one cluster, and the K-means regions on D2, which beat fake data while joining unrelated belts.

3.7 Runtime and scalability

Runtime is compared only where a method reaches ARI > 0.7, which happens on D1 and D6 only; it is one fit's time per 1,000 points at the rule's setting (Table 6). In the case where ARI > 0.9, only one method qualifies on each dataset: Spectral on D1 (14 ms) and GMM on D6 (39 ms). In the case where ARI > 0.7, the ranking from fastest is Spectral (14 ms), then HDBSCAN (18 ms) on D1, and K-Means (26 ms), then K-Means++ (33 ms), then GMM (39 ms) on D6.

Table 6: Methods reaching each ARI level, ranked by fit time (ms per 1,000 points), fastest first.

Data	**ARI** > 0.9	**ARI** > 0.7
D1 ($n = 1{,}000$)	Spectral (14)	Spectral (14), HDBSCAN (18)
D6 ($n = 178$)	GMM (39)	K-Means (26), K-Means++ (33), GMM (39)

On small data a fixed start-up cost dominates, so these times say little about scaling. The largest costs come from model size, not from n: GMM with full covariances in 50 dimensions takes 14.7 s on the 3,000 images of D5, against at most 0.33 s for every other method there. On the 20,000 sales of D8 one fit takes 2.4 s for Spectral on a nearest-neighbour graph, 1.6 s for HDBSCAN, 1.5 s for GMM and 0.6 s for DBSCAN. Choosing the setting costs far more than fitting it: selecting K for K-means, which fits every K with 10 seeds and on 3 fake copies, takes 126 s on D3 and 130 s on D8. A dense RBF affinity for Spectral would need memory quadratic in n, so it is used only on D6; the nearest-neighbour graph is sparse and ran on every dataset.

3.8 Success and failure cases

3.8.1 Successes

Spectral on D1: curved clusters. Shown in Figure 1. Spectral clustering on a 10-nearest-neighbour graph recovers both moons exactly (ARI 1.000). It succeeds because each moon is a chain of close neighbours, and the gap between the moons is wider than any link in the chain. K-Means++ and GMM fail on the same data (ARI 0.481 and 0.501) because they assume convex clusters.

HDBSCAN on D2: belts of varying density. Shown in Figure 2. HDBSCAN returns 28 clusters that follow the plate boundaries, none holding more than 7.5% of earthquakes, with a silhouette of 0.756 against 0.317 on fake data. It succeeds because its density threshold adapts to each cluster, the property D2 needs most; DBSCAN, with one radius, joins the dense western Pacific into one cluster of 35.7%.

GMM on D4 and D6: compact classes that touch. GMM has the best ARI on both (0.618 on D4, 0.915 on D6), ahead of K-Means++ (0.508 and 0.897). The classes are compact groups with no density gap between them, the case centroid and mixture models are built for, and GMM's per-component covariance fits classes of different spread and orientation better than K-means' round clusters.

3.8.2 Failures

DBSCAN on D1: the silhouette picks fragments. Shown in Figure 1. The selection rule picks $\varepsilon = 0.049$, which breaks the moons into 72 fragments with 47.1% noise (ARI 0.009), although the same grid contains a setting with ARI 1.000. Cause: the silhouette scores 72 tight fragments (0.523) above two curved moons (0.387). The fake-data check flags it: shuffled data score 0.822 at the same setting.

Density methods on D3 and D8: one mass. Shown in Figure 4. With no gaps between groups, DBSCAN puts 97.7% of pickups (D3) and 99.7% of sales (D8) in one cluster, and HDBSCAN 91.5% and 99.7%. Cause: no gaps between groups. On D8 the silhouette (0.438, 0.496) and the fake-data check both approve the result; only the largest-cluster share reveals it.

Density methods on D5: high dimensions. DBSCAN and HDBSCAN each return two clusters with 23% to 33% noise and ARI 0.028 and 0.023, while the k-based methods reach 0.34 to 0.40 (Figure 5). Cause: in 50 dimensions an ε large enough to connect points within a class also connects the overlapping classes, and no setting in the grid does better than ARI 0.066.

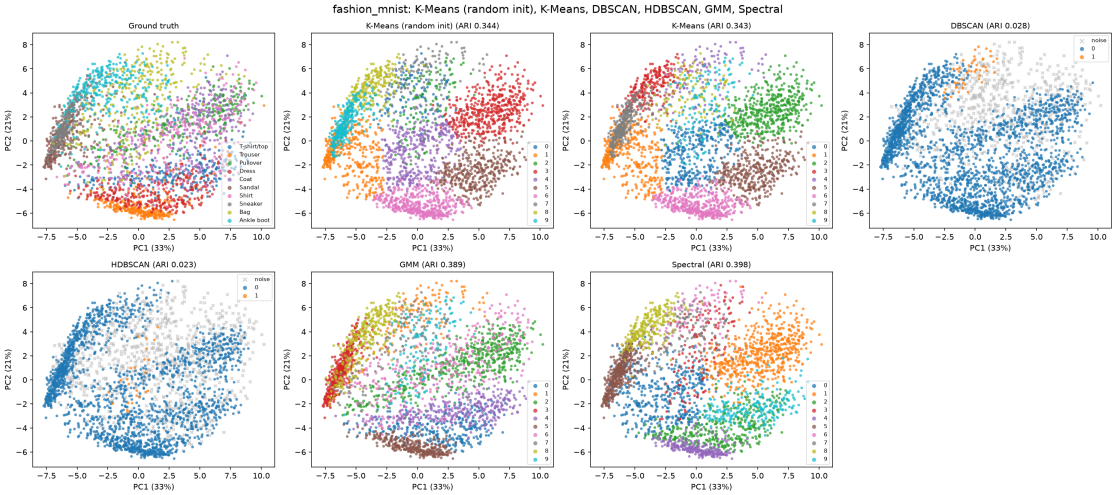


Figure 5: D5 Fashion-MNIST (3,000 images; clustered in 50 PCA dimensions, plotted in the first two). DBSCAN and HDBSCAN merge most classes into one cluster (ARI 0.028, 0.023); Spectral, GMM and K-Means++ reach ARI 0.398, 0.389 and 0.343. Grey crosses: noise.

HDBSCAN on D4: rare classes lost. At the setting chosen without labels HDBSCAN puts 97.0% of proteins in one cluster (ARI 0.038) and still has the highest silhouette on D4 (0.559). Even its best setting by ARI (`min_samples` 10) reaches only 0.398 by calling 29.8% of proteins noise. Cause: a class of 2 to 20 proteins in seven dimensions is not dense enough to form a cluster of its own, so HDBSCAN discards exactly the classes that are hardest to find, and the silhouette, which skips noise, rewards it.

3.9 Key factors and practical guidance

Key factors. Ordered by the size of their effect in our experiments:

1. **Cluster shape against the method's assumption.** On D1 K-Means++ reaches ARI 0.481 and GMM 0.501, against 1.000 for Spectral; on D2 K-means and Spectral draw regions that join unrelated belts (Sections 3.4.1 and 3.4.2). No choice of K repairs a wrong assumption about shape.
2. **Gaps between groups.** Density methods need empty space between clusters. Where there is none they return one dominant cluster: 99.7% of sales on D8, 97.7% of pickups on D3 and 97.0% of proteins on D4 (Sections 3.4.6, 3.4.3 and 3.4.4).
3. **Dimensionality.** On D5, with 50 dimensions, both density methods fall to ARI 0.03 or below, and no setting in their grids exceeds 0.066; the k-based methods reach 0.34 to 0.40 (Section 3.4.5).
4. **Parameter choice without labels.** The rule's choice can be far from the best setting in the same grid: DBSCAN on D1 scores ARI 0.009 against 1.000, HDBSCAN on D4 0.038

against 0.398, and GMM on D4 0.618 at the number of classes against 0.741 at 7 components (Sections 3.4.1 and 3.4.4).

5. **Uneven density.** On D2 HDBSCAN follows the belts with 28 clusters, none above 7.5% of earthquakes, while one DBSCAN radius joins the dense western Pacific into a cluster of 35.7% (Section 3.4.2).
6. **Initialisation.** At the same K and with 10 starts each, K-Means++ matches the labels better than random starts on D4 (ARI 0.508 against 0.379) and is within 0.001 of it on D1, D5 and D6 (Table 4).
7. **Runtime.** It matters only for GMM with full covariances in many dimensions (14.7 s on D5) and for parameter search, which costs more than 100 times a single fit (Section 3.7).

Practical guidance.

1. **Choose by the structure of the data, not by a score.** Plot the data or a projection first. If the groups are curved or long, no choice of k will help K-means; if they are compact and touching, or the data have more than a handful of dimensions, no choice of ε will help DBSCAN.
2. **Do not pick settings by the silhouette alone.** It favours tight fragments (D1 DBSCAN) and settings that call hard points noise. First restrict the search to what the analysis needs (a noise cap, a minimum number of clusters, a cap on the largest cluster), then compare within that range.
3. **Report the noise fraction and the largest cluster next to every silhouette.** A silhouette computed after discarding 42% of the points (D6 HDBSCAN), or with 99.7% in one cluster (D8), says little.
4. **Compare every result against fake data.** Rerun the chosen setting on shuffled columns. It catches results with no real structure (D1 DBSCAN, D4 and D5 HDBSCAN), though not one dominant cluster (D8) or the wrong structure (D2 K-means).
5. **Standardise before K-means, and use K-means++ with several starts.** Random starts lose ARI on D4 even with 10 starts.
6. **Represent coordinates so that distance means something.** Unit vectors on the sphere for global coordinates (D2), a local projection to kilometres within a city (D3); a DBSCAN radius in kilometres can then be checked by someone who knows the area.

4 Conclusion

We compared K-means, K-means++, DBSCAN, HDBSCAN, GMM and spectral clustering on seven datasets, each chosen to test one data property, with every method tuned by the same label-free rule. Density and graph methods beat centroids when clusters are separated by gaps and have any shape, in few dimensions: on D1 spectral clustering recovers the moons exactly where K-means reaches ARI 0.48, and on D2 HDBSCAN follows belts that K-means and one DBSCAN radius merge. Centroid and mixture models win when classes are compact and touch, or the data have many dimensions: on D4 GMM reaches ARI 0.618 and K-Means++ 0.508 while HDBSCAN puts 97% of proteins in one cluster, and on D5 the k-based methods reach 0.34 to 0.40 against 0.03 or below for the density methods. Where the data have no gaps at all, as in the core of D3 and in D8, neither family finds natural groups: the density methods return one cluster holding 92% to 99.7% of the points, and K-means draws borders where its objective puts them.

The silhouette did not tell us which case we were in. On all four labelled datasets the method with the top silhouette was not the one that matched the labels best, and on D1 it chose a DBSCAN

setting that breaks the moons into 72 fragments over one that recovers them exactly. Rerunning each result on fake data helps where a method finds nothing real (D1 DBSCAN, D4 and D5 HDBSCAN), but not where it returns one dominant cluster (D8) or the wrong structure (D2). The choice of method should follow from what the data look like and what the analysis needs, and every score should be checked against fake data, against the points it ignores, and against the share of the largest cluster.

Limitations

[**HoangAnh**] On D6, the methods that need k were given the number of classes, which an analyst would not know. The label-free criterion agrees for K-means (it also picks $k = 3$), but BIC alone would give GMM 4 components, and D6 is small (178 points), so one wine moves a score noticeably.

D3 has no labels, so its clusters are judged against known geography only, and it is one 15,000-point sample of one month. The D4 settings with the best ARI use the labels and overstate what an analyst could reach without them; D4 is also small (336 proteins), so a single protein moves a rare class's share by up to 50%.

D2 and D8 have no labels, so their failures are judged against plate-boundary geography and against what the clusters contain, and cannot be scored. The density parameters are counts of points, so the clusters would change with the size of the catalogue or the sample. The fake data shuffle each column on its own; for the unit vectors of D2 this moves points off the sphere, so the D2 null silhouettes are a rough baseline only.

[**Lam**: one line: datasets go up to 50 dimensions only (D5).]

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